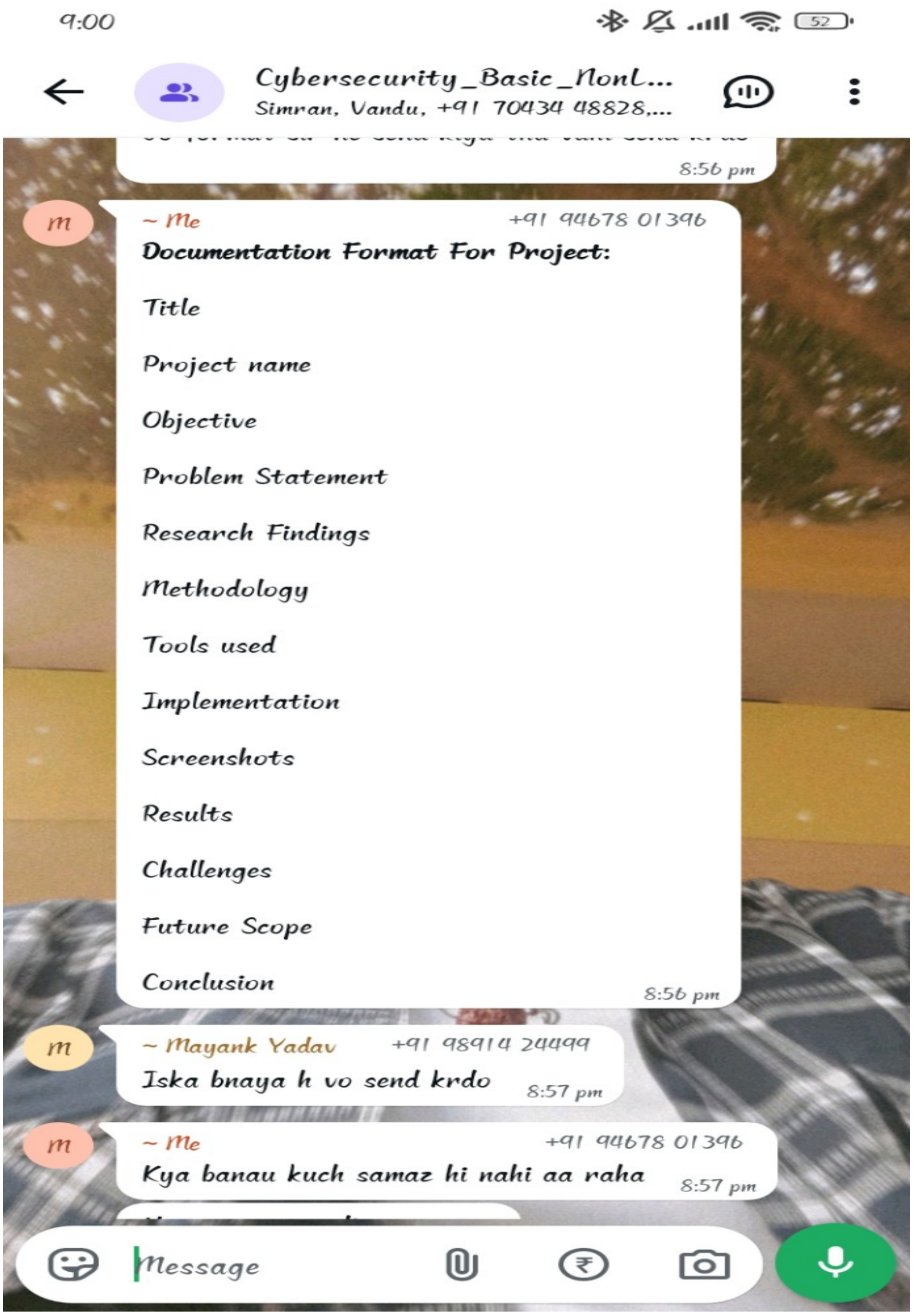

CYBERSECURITY THREAT DETECTION AND ANALYSIS SYSTEM

Project By: Anamika

Title

Cybersecurity Threat Detection and Analysis System

Project Name

CyberShield – A Python Based Cybersecurity Threat Detection and Analysis System

Objective

The objective of this project is to identify suspicious URLs and potential cyber threats using a Python-based analysis system.

Problem Statement

Cyber threats such as phishing attacks, malicious websites and fraudulent links are increasing rapidly. Users often fail to identify suspicious websites before sharing personal information.

Research Findings

Research shows that phishing attacks remain one of the most common forms of cybercrime. Keyword and URL analysis can help identify potentially dangerous websites.

Methodology

User enters a URL. The system checks for suspicious keywords, calculates a threat score and displays a threat level.

Tools Used

Python 3, VS Code, GitHub, Windows OS

Implementation

Python code is used to analyze URLs and classify them into safe, medium-risk or high-risk categories.

Screenshots

Add screenshots of code execution and outputs here.

Results

The project successfully identifies suspicious URLs and alerts users about possible threats.

Challenges

Limited keyword database and inability to detect highly advanced attacks.

Future Scope

Machine Learning, real-time threat intelligence and browser extension integration.

Conclusion

This project demonstrates how Python can be used to build a simple cybersecurity solution for threat detection and awareness.

Python Code

```
suspicious_keywords = ["login","verify","bank","secure","update","password","account"] url =
input("Enter Website URL: ").lower() threat_score = 0 for keyword in suspicious_keywords: if
keyword in url: threat_score += 1 if threat_score >= 2: print("ALERT: Potential Cyber Threat
Detected!") print("Threat Level: HIGH") elif threat_score == 1: print("Threat Level: MEDIUM") else:
print("Website Appears Safe") print("Threat Level: LOW")
```